Project Report

on

TEAM BUILDERS: AI-Based Intelligent Hackathon Team Formation Through TF-IDF Factorization, Collaborative Filtering, and Graph Neural Network Modeling

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that Project Report entitled "AI-Based Intelligent Hackathon Team Formation Through TF-IDF Factorization, Collaborative Filtering, and Graph Neural Network Modeling" which is submitted by "Prachi Rani", "Pari", "Sujal Singh" and "Kartik Singh" in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

The Team Builders platform is designed to provide an intelligent solution for forming well-balanced teams in hackathons and collaborative projects. The system uses artificial intelligence techniques to match participants based on their skills, interests, and experience. It helps users connect with suitable teammates by analyzing their profiles and identifying compatibility.

The platform incorporates AI-based matchmaking methods such as TF-IDF, cosine similarity, and Graph Neural Networks (GNN) to recommend teams that are balanced in terms of skill diversity and collaboration potential. These techniques enable the system to understand both similarity and hidden relationships among participants, leading to more effective team formation.

By automating the team formation process, the system reduces manual effort and improves efficiency. It also supports better collaboration by ensuring that team members have complementary skills and shared goals. The platform is scalable and can be applied in various environments such as hackathons, academic projects, and organizational teamwork.

Overall, this project demonstrates how AI-driven matchmaking can enhance team formation, improve collaboration, and create more productive and successful project outcomes.

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LIST OF ABBREVIATIONS

TFP	Team Formation Problem
GNNs	Graph Neural Networks
GCN	Graph Convolutional Network
GAT	Graph Attention Network
MERN	MongoDB, Express.js, React.js, Node.js
TF-IDF	Term Frequency – Inverse Document Frequency
JWT	JSON Web Token

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Finding suitable teammates with same goals, required technical skills is and big challenge [1]. Team formation problem (TFP), has been studied across different fields like education, software engineering, etc. It requires a bunch of people from bigger bunch of available candidates regarding different skills, experience, personality [2]. GNN, in particular Graph convolution networks (GCN) [3] and graph attention networks GATs [4], can be used to design relation between participants as a graph, where node is a user, and edge between nodes describe a common technical attribute.

These deep learning models are used to provide improved prediction accuracy in complex network environments. Recommendation algorithms like collaborative filtering [5], factorization machines, and graph-based learning can be further upgraded in order to increase the match making accuracy by learning hidden patterns across participants, skills and network behavior. Such up graduation allows evolution of the platform as more user interact with system, this improves team recommendation over time. Dual role access enables participants to register and navigate through hackathons, while admin can create and organize team, this assures fair participation [6].

The complexity of this problem grows together with number of participants and applied constraints. So, it is called NP hard optimization problem [8]. Commonly use methods fail to scale efficiently. This lack of coordination results in unbalanced teams, therefore reducing participation rates and purpose of hackathons [10]. Hackathons are events that promote teamwork, creativity, and problem solving among organizations. They help to form an environment of creativity and teamwork and minimize gap between academic learning and practical implementation [15].

This paper deals with these challenges by presenting TeamBuilders, a MERN web platform that helps in Finding suitable teammates with same goals, required technical skills is and big challenge [1]. Team formation problem (TFP), has been studied across different fields like education, software engineering, etc. It requires a bunch of people from bigger bunch of available candidates regarding different skills, experience, personality [2].

GNN, in particular Graph convolution networks (GCN) [3] and graph attention networks GATs [4], can be used to design relation between participants as a graph, where node is a user, and edge between nodes describe a common technical attribute. These deep learning models are used to provide improved prediction accuracy in complex network environments. Recommendation algorithms like collaborative filtering [5], factorization machines, and graph-based learning can be further upgraded in order to increase the match making accuracy by learning hidden patterns across participants, skills and network behavior. Such up graduation allows evolution of the platform as more user interact with system, this improves team recommendation over time. Dual role automation of the process of forming teams for hackathons.

The system uses the technologies of AI and ML technologies to provide a genius matchmaking of participants for organizing teams. The platform TeamBuilders collects features into its design centered on the user including real time availability of seats, linking of profiles, etc. By focusing on transparency and data driven information, this system helps build an environment that result in innovation and learning.

1.2 PROJECT DESCRIPTION

1.2.1 Problem Statement

Hackathons, and the variety of project based competition formats that are so popular in colleges/universities/organizations, offer an ideal opportunity to gain technical knowledge, creativity, team-work and problem solving skills. Students are able to apply themselves to actual world problems and create unique solutions to those problems under a tight time constraint. One of the most difficult things that many participants face in these types of events, is locating appropriate team members. So many students enter alone and generally don't have relationships with other participants (or the participants' classmates), which means there will be no prior relationships that can provide the necessary technological skill level, interest levels, etc., as well as project objectives. Therefore, many participants spend considerable amounts of time looking for team members rather than developing ideas for their projects and implementing them.

In contrast to most hackathons, where teams are put together either by chance (randomly), through prior knowledge (e.g., through friends), or at the very end with little to no planning; it is typical for such processes to result in unbalanced groups. This means that although all necessary technical skills will likely have been covered by one or more of the team's members, certain skills might still be missing, whereas others will have been duplicated amongst team members. Poor communication, incompatibility and differing expectations among group members also tend to cause significant problems for a team’s productivity as a whole.

The above mentioned issues become increasingly complicated once the amount of participants increases, which then leads to manual finding of ideal teammates being both time consuming and complicated. Most existing hackathon platforms primarily support event registration and organization functions but do not provide the functionality for intelligent team recommendations based upon skills, interests, experience, etc.

1.2.2 Motivation

Many of the students at hackathons and tech competitions have many great ideas and a lot of technical skill. However, it is difficult for them to succeed because there is difficulty in building the right team. Some of these students can spend countless hours looking for teammates that will work out (or be a good fit) for their team. Unfortunately, many times when the time comes to find a team, many students who are relatively inexperienced in hackathons, or lack solid relationships within the community may struggle to find an opportunity to be on a competitive team.

The problem is very frustrating to many of the attendees as it will negatively affect the attendee's desire to attend other similar future events. In addition, the issue has an effect on the overall goal of the Hackathon, which is to provide an environment where there are opportunities to learn, collaborate and innovate. As a result of seeing this problem, we developed a smart platform concept to assist in the team creation process. Our motivation was to utilize Artificial Intelligence (AI) and Machine Learning (ML), so attendees could quickly locate compatible team members and remove the challenges associated with creating teams manually

1.2.3 Objectives of the Project

The overall goal of the Team Builders project is to create a smart platform which will be able to automate the entire process of selecting teams in collaboration with participants in hackathons and other collaborative activities. This platform would help participants rapidly locate suitable teammates using information about the participant's technical abilities, interest in specific project topics, experience levels and compatibility. As a result of being automated, this system would allow participants to spend less time searching manually for teammates, allowing them to concentrate more on producing new creative ideas.

Using machine learning techniques (TF-IDF, Cosine Similarity, Graph Neural Networks) to process participant information in order to recommend quality participants for your team. The purpose of this software is to build well-rounded teams with a balance of skills so that each member has something unique to offer. In addition, the goal of this software is to assist you in

communicating and working effectively with your teammates using tools such as profile development, team suggestions, live updates on projects, etc.

Additionally, the project aims at increasing the number of participants in hackathons through better facilitation of the team-building process. The system will assist organizers with managing both teams and their respective members, as well as other hackathon-related events. Overall, this project intends to build an intelligent, scalable, easy-to-use, team-based application that can improve collaboration for users participating in hackathons.

1.2.4 Scope of the Project

The Team Builders Project primarily focuses on Hackathons, Coding Competitions, Academic Projects, Collaborative Environments (i.e. Teams) that require a high level of Team Formation. Students will use the platform as a way to find teammates that have the appropriate technical skills to complete their project requirements. Colleges and Organizations can also utilize this platform to assist with managing Hackathon Events and enhance Collaboration amongst participants in these events.

Currently, this platform collects participants' personal data (name, email, etc.), assesses their skills and/or interests, generates team recommendations by utilizing machine-learning techniques; in addition, there are many "real-time" functions including an update feature that alerts teams when members are available or unavailable; and messaging tools to enhance the user's ability to interact with others on the site. This project was developed as a highly scalable application using MERN technology stack architecture.

The potential for expansion in terms of the domain of application is significant. For example, it could be applied to all types of collaboration (team formation) within companies; for collaborative research; for freelance projects; etc. The platform could also be linked to social media type applications like LinkedIn and/or GitHub, to enhance the data available from user profiles. As well, advanced technology can be used such as NLP and behavior analysis can increase the accuracy of recommendations made by the system.

CHAPTER 2

LITERATURE REVIEW

Team formation has been widely studied in different fields such as social networks, education, and collaborative work environments. Researchers have developed various models and algorithms to form efficient and balanced teams based on skills, experience, and compatibility among individuals.

Lappas et al. [1] team formation has been an important topic within organizations. it needs a good combination of people who work with coordination. [1] gave one of the oldest models in a proper team selection over organizations by designing problems in a way that aims to organize teams filled with all the required roles along with reducing the communication cost. This helped for computational approaches for team formation, which led to the formation of all other methods that include real world network dynamics.

Building on this, Li et al. [2] increases team structure by using smart algorithms that can find optimal teams with respect to skills required. Their framework is based on practicality - a challenge that TeamBuilders deals with by using databases like MongoDB and Node.js for real time team formation in hackathons. Kipf and Welling [3] introduced GCNs, they are capable of designing relationships between participants using nodes interconnected based on shared skills or previous collaborations.

This way, Graph Attention Networks, as introduced by Veličković et al. [4], improve efficiency to properly capture importance of relationship between different users. These models led the future scope of TeamBuilders in which AI based matching would turn into graph-based analysis of participants to recommend required teammates. Recommendation systems have proven to be good tools for mutual decision making. [5] term - weighing models by Salton and Buckley [6] tells us to how modern algorithms see items for similarity and relativeness. Su and Khoshgoftaar [7] enhanced this more by introducing a survey on collaborative filtering techniques, breaking

them down into memory - based, model based, and hybrid systems. In respect to the TeamBuilders this means skill based filtering and personalized recommendations, this allow participants to find their most compatible teammates.

The work of Ciolfi et al. [8] researched algorithmic models, categorizing them according to the several objective functions, includes skill varieties, and communication potential. The Teambuilders approach ensures balanced team structures that avoid skill overlapping. Anagnostopoulos et al. [9] discuss impact and correlation problems in social networks, in team building scenarios it will determine participants who will likely collaborate based on collaboration patterns of influence.

Yu et al. [10] targeted at team formation in organizations, introducing models that update team assignments with respect to user interaction and requirements. this change organizes the real time seat availability and the team management system in TeamBuilders, making sure users seamlessly join or leave teams based on live events.

Farasat and Nikolaev [11] introduced models that stores team social structures and preferences while stressing the importance of interpersonal relationships and compatibility to technical skills. TeamBuilders works on this principle by enabling users to connect to professional profiles like LinkedIn and GitHub, this makes sure that when recommendations are shown then both skills credibility and collaboration are considered.

[12] factorization-based models like Rendle's Factorization Machines [13], allows for making more advanced and predictive recommendation engines. Gamification has also been seen as a major tool of improving engagement within organizations.

Rahman et al. [14] performed a survey in which it was researched that how gamification elements like badges, leaderboards, and progress improve motivation to participate. Adding such elements within TeamBuilders makes sure that participants don't leave in hackathons and makes participation more meaningful.

[15] the main challenges: lack of visibility, skill mismatching. These findings make main purpose of TeamBuilders, which we will try to solve these issues through a web platform, AI - assisted to help whole process of finding, forming and managing teams in hackathons.

CHAPTER 3

PROPOSED METHODOLOGY

It uses a DDR approach, where the technology is built and then checked with users. A working prototype was shared with college students and hackathon participants. The system was developed with the MERN stack to provide a scalable platform and real-time communication for hackathon participants. They made profiles where they added their skills and what they prefer. The AI module provided recommendations for team formation. Participants gave feedback that helped improve the system's usability and accuracy.

The AI module uses TF IDF, cosine similarity, and a GNN model to evaluate participant compatibility. The GNN represents users as nodes in a network and learns patterns such as matching skills. Quantitative data was analyzed using Python with pandas and NumPy to measure matching accuracy, response time. Qualitative feedback was used to understand user experiences and the reliability of the system.

3.1. SYSTEM ARCHITECTURE

The system is built on a modular and scalable architecture designed to support smooth interaction among participants, administrators, and the AI-driven matchmaking engine. Overall system architecture showing frontend, backend, and database layers. The architecture consists of three core layers: the frontend, backend, and database, all connected through secure RESTful APIs to maintain consistent communication and efficient data flow across the platform.

The frontend layer is written using React.js and allows access to an interactive dashboard where new users are able to signup, see events, and change their profile. The users can also form or join teams. The user-friendly interface can also be updated in real-time like new team requests and event progress. The backend was picked as Node.js and the express which will deal with the user authentication, session management and all the business logic in the form of REST APIs.

The system which verifies the formation of the teams based on some criteria in which the number of team members is limited and members are assigned roles where they belong falls

under the rule engine. It should also have the backend enabled with smart match making system that proposes teams on the basis of compatibility. MongoDB is chosen as the data layer on which the details about the participants of the event, skills, information about teams, event data, and so on are going to be stored. The flexible schema of the same can easily expand to accept the addition of new fields such as LinkedIn, GitHub, or achievements and the extremely fast queries will ensure system responsiveness on a timely basis.

The matchmaking module is an important part of the system based on AI. It provides compatibility scores because it considers the skills, experience as well as preferences of each user. The system does not use simple similarity measures, but uses a GNN to demonstrate the similarities between the participants. In this model, the users are managed as each node, and common skills or prior co-operation are the connections between them.

In this way, the system generates recommendations of teams considering technical competencies and the possibility of the participants to cooperate with one another. The basis of these recommendations is on the potential of good workmanship in any team and is not just limited to the capabilities that match the job but rather to the propensity of the capability working in any team. During the future employment, the system may be improved with sentiment analysis and simple personality test that, consequently, will guarantee even higher staffing and total productivity.

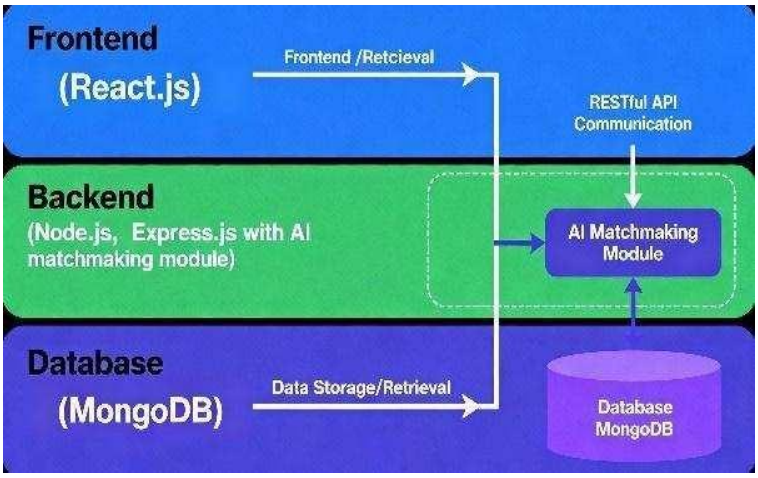

Fig.1. Overall system architecture showing frontend, backend, and database layers.

3.2. IMPLEMENTATION

The project is implemented in a mobilized manner in which all modules interact. All modules are particular to make the system convenient and interactive, as well as helpful to students desiring to create teams, compete, and collaborate effectively. The design is geared towards a user-friendly design, which can expand as more users are added, and which can easily coordinate among additional functionalities. Consequently, the whole system becomes one, well linked platform that is simpler and convenient to the users. The first to be introduced will be a User Authentication and Profile Creation that will be highly secure.

The learners will be in a position to log in by using their email and password. The platform retains the login sessions through JWT to improve the security of the data and the proper handling of the login sessions. Once this, the user will be able to establish profiles as per their preferences, which means to show their technical skills, preferred positions, and areas that they wish to receive assistance. It is a systematic profile creation which becomes the base of AI-generated recommendations at a subsequent point. The main part of the system is the Skill Matching and AI Recommendation Engine.

This module uses TF-IDF and cosine similarity to check how well participants match based on their skills, and project interests. The system also looks at how users usually work together and their past collaboration behavior to calculate a better score. Based on this analysis, the module suggests suitable team members and projects, helping students connect with others who have another skill. Its primary component is the Skill Matching and AI Recommendation Engine presented.

This module makes use of TF-IDF and cosine similarity to verify the level of matching of the participants of different areas of skills, and the project interests. The system also examines the way the users tend to collaborate and how they have behaved historically as a team in order to derive an improved score. Upon the analysis, the module proposes an appropriate team and project, and assists the students to establish contact with other students possessing another skill.

The Project Navigator and Role Mapping module will be used to support the different contests and projects, which produces a graph-based mapping using decision tree logics that can in turn

be used to match the individual strengths of each person with the project needs and milestones. It also provides full insight into the role of every member of a project hence ensuring that the contribution is relatively balanced throughout the entire project lifecycle. Other elements of the system are included to encourage students to participate actively with the help of progress tracking and basic gamification. The students get points and badges to participate and on the other hand there are leader boards to encourage greater interest and healthy competition. One of the examples of a heatmap like that at GitHub indicates the amount of activity every day, letting teams monitor what they are accomplishing and how it is being achieved. The features ensure that the platform is more interactive and helpful to students.

The final component of the system is Collaboration and Communication module that assists team members in being able to work harmoniously. It enables real-time chat, file sharing, as well as periodic updates on the project activities. Dependable communication APIs are employed in a way that the user is maintained in touch and informed without relying on other external tools. Regardless of whether students are talking about ideas, exchanging documents, or delegating tasks, all of this occurs within a single location. All these modules, together, form a well structured system which will assist a student in profile making to the successful execution of projects.

3.3. MACHINE LEARNING INTEGRATION

3.3.1.Skill and Profile Representation using TF-IDF:

The technical and behavioral characteristics of participants are converted into formats through a text vectorization process, which serves as the foundation of the matchmaking engine. The skills, tools and interests listed by each participant are represented as dimensional feature vectors through TF-IDF . Within the user profile d for a term t the TF-IDF weight is computed as:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log\left(\frac{N}{\text{DF}(t)}\right)$$

Within this context TF(t,d) signifies the frequency of term t in document d N stands for the number of user profiles and DF(t) refers to the number of profiles containing the term t. Each of the profile vectors is compared using cosine similarity, defined as:

$$\text{Similarity}(A, B) = \frac{u \cdot v}{\| u \| \quad \| v \|}$$

where u and v represent the TF–IDF vectors for participants A and B. This produces a similarity score between 0 and 1 enabling straightforward compatibility assessment. For example a participant, with "Python, Flask, Machine Learning" would show similarity to another participant listing "Python, Data Science, Flask " thus improving the contextual clarity and relevance of matches.

3.3.2. Collaborative Recommendations:

Filtering for Personalized Alongside text-based matching TeamBuilders utilizes Collaborative Filtering to uncover connections derived from past data. CF detects patterns, in participants previous teamwork, profile engagements and team choices; thus, the system is able to suggest teammates even when there is little direct skill similarity. The model utilizes matrix factorization to predict a participant's propensity to work with another person. $R(u, i) = \mu + b_u + b_i + q^T p_u$ where R(u,i) indicates the estimated compatibility, between user u and teammate i, stands for the average, and are bias components and represents latent factors capturing hidden user team relationships. Moreover, factorization machines can be utilized to handle sparse feature matrices while capturing complex pairwise feature interactions. In addition to ensure user's data expansion, it is scalable, reliable as per events.

3.3.3. Graph Neural Network Modeling:

Graph Neural Network (GNN) is utilized as per the framework for modelling and forecasting team compatibility among team members. We represented participants and their interactions on a graph structure efficiently both characteristics and relationships in between participants. Nodes are representing individuals within the graph and the links or edges between these nodes that carry weighted data reflecting skills intersections, domains and previous collaborations, and previous experiences among the participants. This allows the model to go beyond tabular connections and grab , dynamically and effects within a team's network. Each node the graph is associated with a feature vector $x_i = ["skills, experience, domain, collaboration\ score"]$. The

technical profiles of every participants joins these characteristics. The e_ij edges links the nodes and are established based on 2 criteria: cosine similarity to participants skill vectors, which evaluates the match of their own technical abilities and the count of part interactions or collaborations are among participants indicating their history teamwork. The result, consequently being captures the expertise distribution and the relational dynamics among users forming an insightful teamwork, for learning. The GG suggestions enables the dissemination and accumulation of features, across graph. The GCN functions can be precise; y expressed as:

$$H^{(l+1)} = \sigma(\widehat{D}^{-1/2}\hat{A}\widehat{D}^{-1/2}H^{(l)}W^{(l)})$$

where D represents the associated degree matrix, H(l) stands for featuring matrix at the I-the layer W(l) denotes a notable weight matrix is a nonlinear activation such as ReLU and A = A + I is adjacency matrix of the graph, with self-loops added that ensuring each node incorporates its own features during the update process. This method of propagation allows the model to progressively collect information from every node and refine it until the representation of every participant is context-sensitive, by shaping the attributes of the participants they link to. By this way the initial GCN layer collects information from the neighbors reflects how direct connected participants affect each others. Effectively uncovering more complex relationships, within the analyzed network of users. Consequently, models can be identify crucial patterns, how skill variety and experience complement each other to facilitate the successful collaborations. For each and every pair or group of individuals, the GCNs linked output layer forecasts a compatibility score to normalized between 0 and 1.

This is representing the probability that a particular arrangement would collaborate effectively to reach to high performance. The Mean Squared Error loss function is employed to evaluate the difference between the network's score compatibility and actual performance outcomes, this is throughout the training. GNN captures a representation of the core elements influencing the alignment and experience diversity balance by steadily reducing this loss alongside interpersonal cooperation metrics in effective teamwork of user. Summarizing the project will utilize the Graph Neural Network model as mechanism, for team creation or formation. Retention of Users:

Throughout the 3-week trail period of a college based event, user's involvement was tracked to assess the engagement levels on different parameters. Using dashboards direct suggestion and teamwork monitoring, by teambuilders participants interaction was steadily upheld. 65% revealed increment in active profile visits, each growing by 1.8 times. Furthermore, the completion rate of creating or forming team may improved from 56% when done traditionally to 84% with the use our Team building website. These stats shows that AI-driven recommendations boosted user's involvement while reducing the time period required for assembling a team.. Recommendation Suitability and Team Composition: To determine the relevance of AI-generated suggestions the teams created on the platform were analyzed for diversity in skills and overall balance.

To prevent any team from being excessively uniform or skewed factors such as experience, interests and skill sets were reviewed. Mentors and faculty evaluators gave the team composition a rating of 4.4 out of 5. They emphasized that the team's roles were more evenly distributed and that the members' skill sets were more varied. Users added that when the team's makeup was determined by both behavioral and technical compatibility, meaningful collaboration tended to occur more readily.

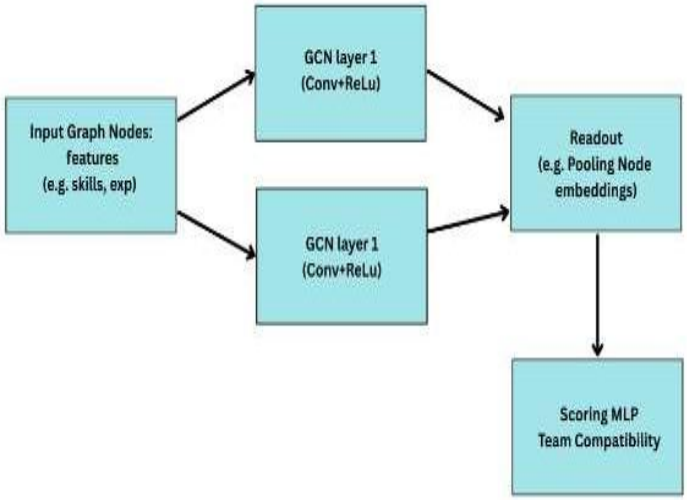

Fig. 2. Model architecture of the proposed GCN-based compatibility predictor.

CHAPTER 4

RESULTS AND DISCUSSION

The evaluation of our TeamBuilders website involved a structured performance on 80 students from various engineering traits who have applied this platform for team formation according to their need and project-based events.

4.1. Team Match Accuracy:

The effectiveness of the system was validated by comparing team suggestions produced by the AI system with expert teams and manually assembled teams. The Graph Neural Network (GNN) model takes into account participant attributes, such as skills and previous experience and interests in a given field, as well as latent connections, such as prior collaboration. This enables the system to provide more accurate and relevant team recommendations than conventional approaches, which are based on limited or subjective data.

Our experiments demonstrate that the system reached an 87% match accuracy, meaning that most of the team suggestions were well matched. Furthermore, the system significantly improved over traditional team formation by almost 20%, showing its capacity to understand latent relationships and form better teams. In short, the model guarantees well-balanced and efficient teams, which enhance teamwork and allow for more accurate and efficient team formation.

4.2. User Engagement and Retention:

In the trial event in the college over a period of weeks, we tracked user engagement with the system. Critical system features included intuitive dashboards, profile-based recommendations and collaboration statistics. These features ensured that the participants were engaged and kept using the system regularly.

We found that 65% of participants were active daily, with activities including viewing profiles, sending friend requests and sending messages. This suggests that the platform was able to keep participants engaged and facilitate ongoing interaction.

Moreover, the success rate of team formation was enhanced. The completion rate of manual team formation was approximately 56%, while the rate when using Team Builders was 84%. This shows that AI-based recommendations, in addition to increasing participation, also save time and effort in forming balanced and productive teams.

4.3. Recommendation Relevance and Team Balance:

The teams that would be formed later via the platform required analysis regarding the diversity and balance of the skills to decide whether the recommendations suggested by the AI were relevant. Their professional experience, interests and skills were examined to ensure each team would not be in a very unbalanced position. The average result of the mentors and evaluators on the quality of the team was 4.4/5. They pointed out that the diversity of skills of the members was increased, with the roles within the team being more balanced. The user responded that the collaboration was more likely to occur when the team composition was based on technical and personal compatibility.

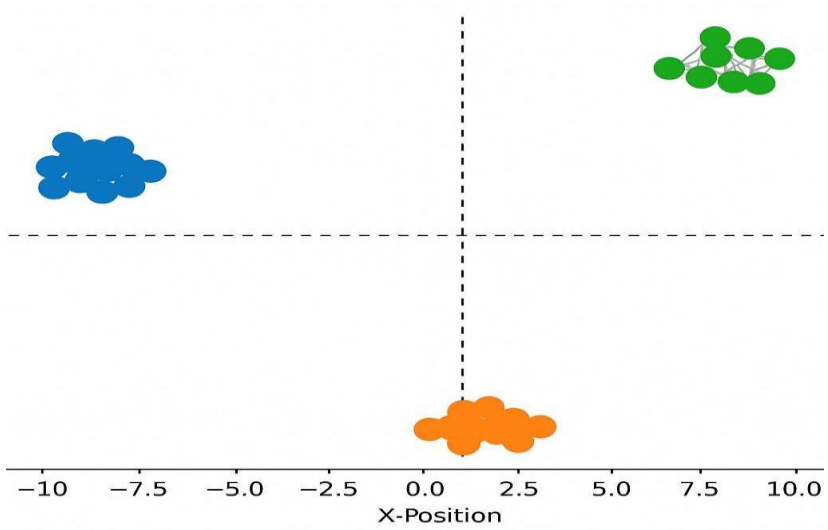

Fig.3. Shows latent skill clusters and relationships

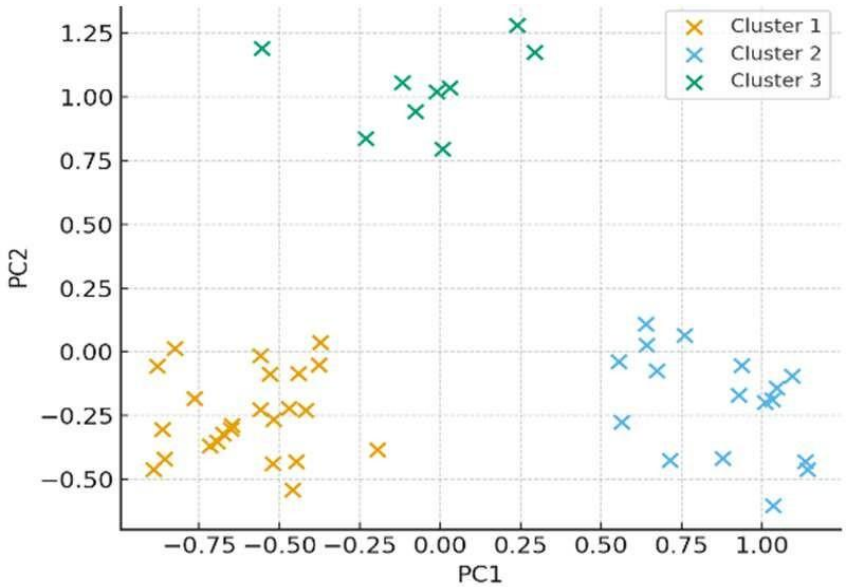

Fig.4. Visualizes learned feature separation and skill compatibility

4.4. Performance Metrics and System Efficiency:

The Team Builders system is assessed using objective measures and user feedback to ensure precise and effective team matching. The system had a matchmaking accuracy of around 87%, indicating that it effectively finds suitable participants using TF-IDF, cosine similarity, and Graph Neural Networks.

The success rate of team formation goes from 56% (manual approach) to 84%, which shows that it is much easier to find teammates using our system. Additionally, the system offers quick response times, allowing for real-time recommendations and user-friendly experiences. Regarding user retention, approximately 65% of users stayed engaged, indicating the system's success in engaging users.

In summary, the system is efficient, scalable and reliable, offering an effective approach to smart and automatic team formation for hackathons and team-building activities.

Metric	Performance Value
Match Accuracy	87%
Precision	85%
Recall	84%
F1-Score	84.5%
Average Response Time	1.9 seconds
Server Uptime	99.1%
Database Query Success Rate	98.5%

Table.1. Performance Metrics

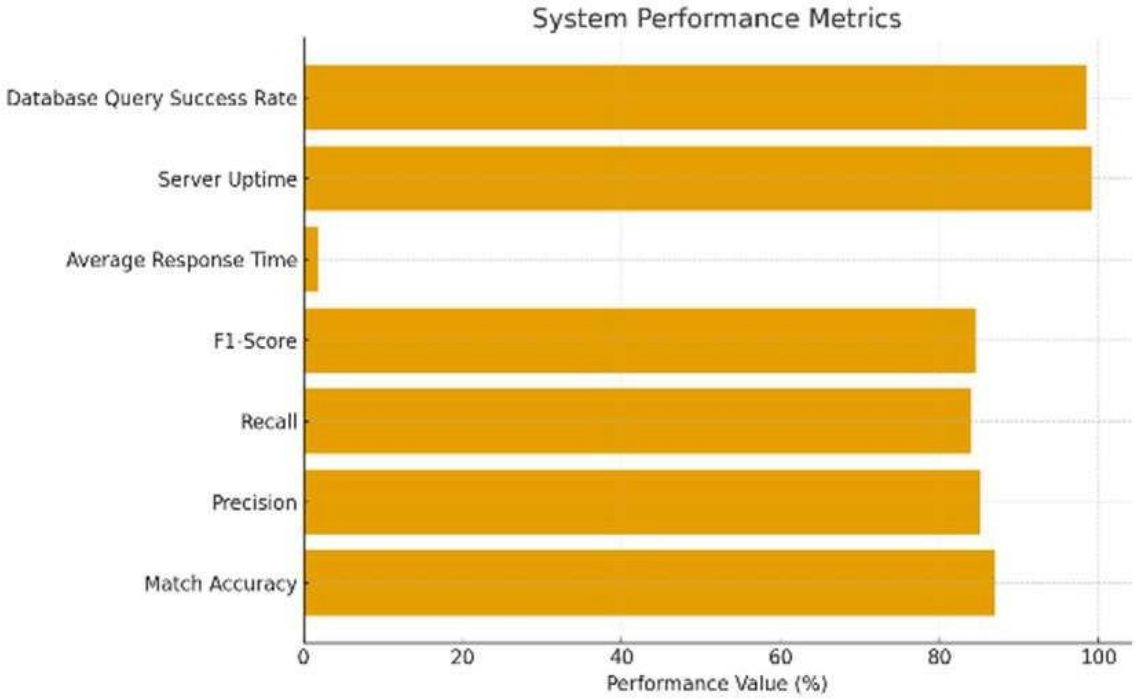

Fig.5. System Performance Metrics

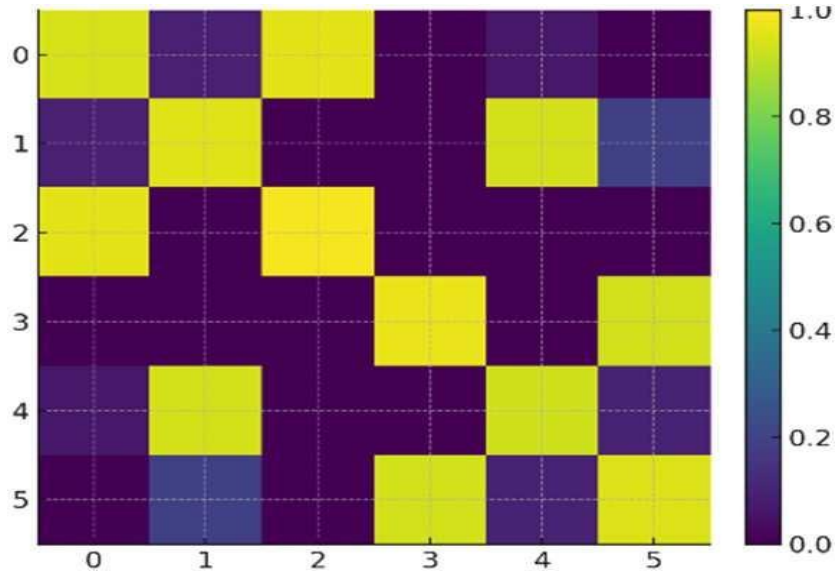

Fig.6. Represents pairwise similarity and validates quantitative results

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The Team Builders system presents an effective solution to the problem of team formation in hackathons and collaborative environments. It addresses one of the most common challenges faced by participants, which is finding suitable teammates with compatible skills, interests, and goals. By using artificial intelligence techniques such as TF-IDF, cosine similarity, and Graph Neural Networks, the system is able to analyze participant profiles and recommend teams that are both balanced and efficient. This approach not only improves the accuracy of team formation but also ensures that each team has a proper mix of skills and roles, leading to better collaboration and performance.

The platform also enhances transparency and decision-making through visualization techniques like scatter plots, which help users and administrators understand the relationships and compatibility between participants. Unlike traditional manual methods, which are often time-consuming and unreliable, the Team Builders system automates the entire process and provides real-time recommendations. This reduces effort, saves time, and increases participation by making the process more accessible and user-friendly.

Overall, the system successfully demonstrates how data-driven and AI-based approaches can improve teamwork and productivity. It creates an environment where participants can collaborate more effectively, leading to better outcomes in hackathons, academic projects, and professional settings. The results show that intelligent matchmaking can significantly enhance the quality of teams and the overall experience of users.

5.2. FUTURE SCOPE

The Team Builders system has strong potential for further improvement and expansion in the future. One of the key areas of enhancement is the integration of advanced deep learning models, which can improve the accuracy and adaptability of the matchmaking process. By using more sophisticated models, the system can better understand complex relationships between participants and provide more refined recommendations.

Another important improvement is the use of Natural Language Processing (NLP) to analyze user profiles, resumes, and previous project experiences in more detail. This will allow the system to gain deeper insights into each participant's abilities and preferences, leading to more accurate compatibility predictions. Additionally, incorporating real-time feedback mechanisms can help the system learn continuously from user interactions and improve its recommendations over time.

The platform can also be expanded by integrating it with existing systems such as university portals, LinkedIn, and GitHub, making it easier for users to import their profiles and increasing the reach of the system. Future developments may also include analyzing behavioral and communication patterns to ensure not only skill compatibility but also better teamwork and coordination among participants.

Furthermore, the system can be adapted for use beyond hackathons, such as in corporate team formation, research collaborations, and online learning platforms. With these enhancements, the Team Builders platform can evolve into a comprehensive solution for intelligent, data-driven team formation across various domains.

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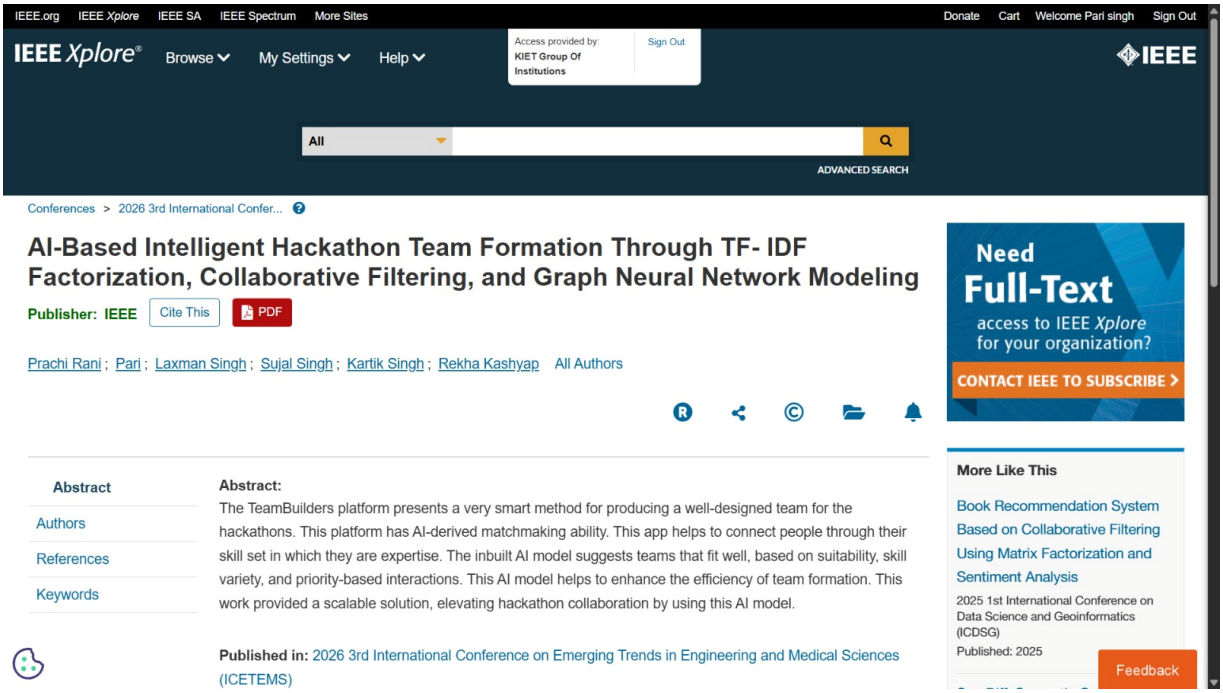

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The TeamBuilders platform presents a very smart method for producing a well-designed team for the hackathons. This platform has AI-derived matchmaking ability. This app helps to connect people through their skill set in which they are expertise. The inbuilt AI model suggests teams that fit well, based on suitability, skill variety, and priority-based interactions. This AI model helps to enhance the efficiency of team formation. This work provided a scalable solution, elevating hackathon collaboration by using this AI model.

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